E-commerce Revenue Prediction using Deep Learning Models (ANN, CNN, LSTM)

Introduction

In today's digital era, e-commerce platforms generate a massive amount of data through customer transactions, product sales, and user interactions. This data contains valuable insights that can help businesses improve their performance, optimize operations, and increase profitability.

One of the most important aspects for any business is **revenue prediction**. Accurate prediction of revenue helps organizations in decision-making, inventory management, demand forecasting, and financial planning.

Traditional machine learning models are useful, but they often fail to capture complex non-linear relationships in large datasets. Deep Learning models, on the other hand, are capable of learning complex patterns and dependencies in data.

In this project, we apply three deep learning models:

- Artificial Neural Network (ANN)
- Convolutional Neural Network (CNN)
- Long Short-Term Memory (LSTM)

These models are used to predict the **Revenue** generated from e-commerce transactions.

☞

"In this project, Revenue is used as the target variable, meaning it is the value we aim to predict. The models are trained using various input features to learn patterns and estimate the expected revenue for new data. This helps in understanding business performance and supports better decision-making."

Overall Objectives

The main objectives of this project are:

1. To understand and preprocess the e-commerce dataset
2. To build deep learning models (ANN, CNN, LSTM)
3. To train the models using the dataset
4. To predict revenue based on input features
5. To evaluate and compare the performance of different models
6. To analyze which model is most suitable for tabular data

Dataset Description

The dataset used in this project is an **e-commerce sales dataset** containing information about transactions.

📌 **Type of Dataset:**

- Structured (Tabular Data)

📌 **Target Variable:**

👉 **Revenue**

This is the output variable that the model tries to predict.

📌 **Input Features:**

These include various attributes such as:

- Product information
- Category
- Region
- Sales-related features
- Customer or transaction-related data

These features act as independent variables used to predict revenue.

Data Preprocessing

Before applying deep learning models, data preprocessing is essential to ensure accuracy and efficiency.

◇ 1. Handling Missing Values

Missing values were removed using:

data = data.dropna()

This ensures the dataset is clean and consistent.

◇ 2. Encoding Categorical Variables

Since machine learning models cannot process text data, categorical variables were converted into numerical form using Label Encoding.

Example:

LabelEncoder()

◇ 3. Feature Scaling

Feature scaling was applied using StandardScaler to normalize the data.

This helps:

- Improve model performance
- Speed up training
- Avoid bias due to large values

◇ 4. Train-Test Split

The dataset was divided into:

- 80% training data
- 20% testing data

This helps evaluate model performance on unseen data.

Deep Learning Models

1) Artificial Neural Network (ANN)

📌 How ANN is applied on this dataset:

1. The dataset contains tabular data (rows and columns).
2. Each row represents one transaction.
3. All input features (except Revenue) are used as inputs.
4. The model takes these inputs and learns patterns to predict Revenue.

📌 Input to ANN:

- Product
- Category
- Region
- Quantity
- Price

📌 Output:

👉 Predicted Revenue value

📌 How model works in this project:

- Data is passed into the ANN model
- ANN processes data through multiple hidden layers
- Each layer learns relationships between features
- Final layer gives predicted revenue

📌 Predicts:

👉"ANN predicts the expected revenue for each transaction based on input features."

📌 Why it works best here:

- Dataset is structured (tabular)
- ANN is designed for this type of data

2) Convolutional Neural Network (CNN)

📌 **How CNN is applied:**

Even though CNN is mainly used for images, in this project:

1. Tabular data is reshaped into a 3D format
2. Each row is treated like a small "signal"
3. CNN extracts patterns between features

📌 Input to CNN:

👉 Same features as ANN, but reshaped:

X_train.reshape(samples, features, 1)

📌 Output:

👉 Predicted Revenue

📌 How model works in this project:

- Convolution layer scans input features
- Detects patterns between variables
- Flattens output
- Fully connected layer predicts revenue

📌 Predicts:

👉 **"**CNN predicts revenue by extracting hidden patterns between input features."

📌 Important understanding:

- CNN is not ideal for tabular data
- Used here only for comparison

3) Long Short-Term Memory (LSTM)

📌 How LSTM is applied:

1. Data is reshaped into sequence format
2. Each row is treated as a sequence
3. LSTM tries to learn dependencies between features

📌 Input to LSTM:

👉 Same features reshaped into sequence:

X_train.reshape(samples, features, 1)

📌 Output:

👉 Predicted Revenue

📌 How model works in this project:

- LSTM processes input step-by-step
- Maintains internal memory
- Learns relationships between feature values
- Outputs predicted revenue

📌 **Predicts**:

👉 "LSTM predicts revenue by learning relationships in sequential form of input features."

📌 Important understanding:

- LSTM is best for time-series data
- Our dataset is not time-based

  👉 So performance is lower

FINAL COMPARISON

Model	How Used	What It Predicts	Performance
ANN	Direct tabular input	Revenue	Best
CNN	Reshaped feature patterns	Revenue	Moderate
LSTM	Sequence form of features	Revenue	Lower

👉

"In this project, all three models (ANN, CNN, LSTM) are used to predict the Revenue generated from e-commerce transactions based on input features such as product details, category, and other attributes."

INTERPRTATION

"In this project, revenue is predicted using ANN, CNN, and LSTM models, where ANN achieved the best performance with the lowest error. This shows that ANN is more effective in learning relationships between input features and revenue for structured (tabular) data. The model captures complex patterns between variables such as product, category, and region to make accurate predictions. Higher predicted revenue indicates strong-performing combinations of features. The results also show that CNN and LSTM are less suitable for this dataset type. However, the model predicts revenue only and cannot determine profit as cost data is not available."